UE26CY151A Engineering Chemistry · Unit 2 · Electrochemistry and Corrosion

Important points

Definitions, statements, key points, "why" reasoning and exam traps, topic by topic.

Class 003 · Types of electrodes · compiled 17 October 2026

ESA ★★★	Asked repeatedly in past ESA papers
ESA ★★	Asked in past ESA papers or computed in class-work problems
ESA ★	Conceptual: taught in class, not examined so far
ADDED FOR CLARITY	Standard context that is not in the class material
TRANSCRIPT ONLY	Said in the lecture but not on any slide
VIDEO ONLY	Seen only in a frame of the class video
NOT ON CURRENT SLIDES	From older material or past papers, not this year's slides
S003·5	Deck 003, slide 5 (rendered .pptx)
T003 00:07:50	AV-summary transcript 003, segment start
PDF p.57	Older consolidated Unit 2 slide PDF (8enterprise), page 57
ESA Dec-23 2d	Past paper: ESA Dec 2023, question 2.d
ESA Jan-25 s1 2b	Past paper: ESA Jan 2025 set 1, question 2.b

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1 Types of electrodes: the map

Definitions

Electrode (half-cell). One of the two halves of a cell: a conductor in contact with an electrolyte at whose interface a potential develops. "In order to form a cell, 2 half cells or 2 electrodes are required."

S003·3 · T003 00:00:31

Statements in the class material

> *"In order to form a cell, 2 half cells or 2 electrodes are required. Various types of electrodes are available which are constructed based on the application."* S003·3

Key points

1. Always write the electrode reaction as a **reduction**, because electrode potentials are reduction potentials; the Nernst log term is then products over reactants. ESA ★★ T003 00:01:38 · T003 00:07:50
2. Pure solids and pure metals have activity 1 and drop out of the log term; gases enter as partial pressure (atm); amalgams stay in. ESA ★★ T003 00:11:06 · T003 00:08:07
3. 0.0591 in the equation means room temperature (298 K): $\mathbf{2.303}RT/F$. ESA ★ T003 00:05:12
4. Three groups: the electrode metal reacts (metal–ion, amalgam, metal–insoluble salt); an inert Pt only carries electrons (gas, redox, quinhydrone); a membrane exchanges ions (ion-selective). ESA ★ ADDED FOR CLARITY S003·3 · S003·9

2 Metal–metal ion electrode

Definitions

Metal–metal ion electrode. Metal in contact with a solution of its own ions, e.g. $\mathrm{Zn/Zn^{2+}}$, $\mathrm{Cu/Cu^{2+}}$, $\mathrm{Ag/Ag^{+}}$. S003·3 · T003 00:00:59

Key points

1. Examples: $\mathrm{Zn|Zn^{2+}}$, $\mathrm{Cu|Cu^{2+}}$, $\mathrm{Ag|Ag^{+}}$; reaction $\mathrm{M^{n+} + ne^- \rightleftharpoons M}$. ESA ★ S003·3 · T003 00:01:13
2. Increasing $[\mathrm{M^{n+}}]$ tenfold raises E by $0.0591/n$ V. ESA ★ added for clarity S003·3

Why?

Why is there a + sign in $E = E^\circ + \frac{0.0591}{n}\log[\mathrm{M^{n+}}]$, when the others have −?
For $\mathrm{M^{n+} + ne^- \rightleftharpoons M}$ the log term is $\log\frac{1}{[\mathrm{M^{n+}}]} = -\log[\mathrm{M^{n+}}]$; the two minus signs make +. Every electrode follows the same rule E1.1.

added for clarity S003·3 · T003 00:02:15

3 Metal–metal insoluble salt–anion electrode

Definitions

Metal–metal insoluble salt–anion electrode. These electrodes consist of a metal in contact with a sparingly soluble salt of the same metal dipped in a solution of soluble salt of the same anion, e.g. calomel electrode $\mathrm{Hg/Hg_2Cl_2/KCl}$, $\mathrm{Ag/AgCl(s)/HCl}$. S003·4 · T003 00:02:31

Key points

1. The electrode reaction product is Ag, not $\mathrm{Ag^+}$: $\mathrm{AgCl + e^- \rightleftharpoons Ag + Cl^-}$. With $\mathrm{Ag^+}$ the equation would not even balance. ESA ★★ T003 00:04:20
2. The solution must contain the anion of the sparingly soluble salt ($\mathrm{Cl^-}$ from KCl or HCl), so the potential depends on $[\mathrm{Cl^-}]$: these electrodes are reversible to the anion. ESA ★★ T003 00:04:50 · PDF p.33
3. Construction of $\mathrm{Ag/AgCl}$: a silver wire coated with AgCl dipped in KCl (or HCl), with a porous junction. ESA ★ S003·4 · T003 00:04:34
4. Both common reference electrodes (calomel and $\mathrm{Ag/AgCl}$) are of this type; class 004 covers them. ESA ★ S003·4 · T003 00:18:34

Exam traps

Exam trap. Writing $\mathrm{AgCl + e^- \rightleftharpoons Ag^+ + Cl^-}$ is wrong (and unbalanced in charge); the product is metallic Ag. T003 00:04:20

Exam trap. The Ag/AgCl and calomel potentials fall as $[\mathrm{Cl^-}]$ rises (minus sign); students often write $+$. (added for clarity) S003·4

4 Gas electrode

Definitions

Gas electrode. It consists of gas bubbling about an inert metal foil, immersed in solution containing ions to which the gas is reversible. The metal provides electrical contact and facilitates the establishment of equilibrium between the gas and its ions, e.g. hydrogen electrode $\mathrm{Pt/H_2/H^+}$, chlorine electrode $\mathrm{Pt/Cl_2/Cl^-}$. S003·5 · T003 00:05:23

Statements in the class material

> *"The metal provides electrical contact and facilitates the establishment of equilibrium between the gas and its ions."* S003·5

Key points

1. Pt is chosen because it is inert; platinised (platinum black) foil gives a large surface for the gas–ion equilibrium. ESA ★★ S003·5 · T003 00:06:07
2. The gas must be bubbled into a solution of the ions to which it is reversible: $\mathrm{H_2}$ into $\mathrm{H^+}$, $\mathrm{Cl_2}$ into $\mathrm{Cl^-}$. ESA ★★ T003 00:06:45 · T003 00:07:03
3. SHE: $\mathrm{H_2}$ at 1 atm, $[\mathrm{H^+}] = 1$ M (1 M HCl), so $E = E^\circ = 0$ V; it is the primary reference. ESA ★★ S003·5 · T003 00:08:38
4. Hydrogen electrode: $E = -0.0591\,\mathrm{pH}$ at 1 atm; it becomes more negative by 0.0591 V per pH unit. ESA ★★ (added for clarity) S003·5

Why?

Why use a partial pressure for hydrogen?
$\mathrm{H_2}$ is a gas; the activity of a gas is its partial pressure in atm. So the log term is $p_{H_2}/[\mathrm{H^+}]^2$.
T003 00:08:07 · T003 00:08:20

Why is the metal in a gas electrode platinum?
It must not react; it only conducts electrons and catalyses the gas–ion equilibrium on its (platinised) surface.
S003·5 · T003 00:06:25

Exam traps

Exam trap. Forgetting the square: the log term is $p_{H_2}/[\mathrm{H^+}]^2$, not $p_{H_2}/[\mathrm{H^+}]$; with $n = 2$ the pH term becomes $0.0591\,\mathrm{pH}$, not $0.0591\,\mathrm{pH}/2$. T003 00:08:20

5 Amalgam electrode

Definitions

Amalgam electrode. It is similar to the metal–metal ion electrode, in which a metal amalgam is in contact with a solution containing its own ions, e.g. lead amalgam electrode $\mathbf{Pb-Hg/Pb^{2+}}$. S003·6 · T003 00:08:59

Amalgam. An alloy of a metal with mercury. transcript only T003 00:08:59

Key points

1. Why amalgamate? Sodium in $\mathbf{Na^{+}}$ solution reacts vigorously, so no potential can be measured; sodium amalgam in $\mathbf{Na^{+}}$ works, and the mercury content controls the activity. ESA ★ transcript only T003 00:10:14 · T003 00:10:36
2. The amalgam's activity depends on the percentage of lead, so $[\mathbf{Pb-Hg}]$ is kept in the log term. ESA ★ T003 00:11:24 · S003·6

6 Oxidation–reduction electrodes and the quinhydrone electrode

Definitions

Oxidation–reduction (redox) electrode. It consists of an inert metal such as platinum immersed in a solution containing an appropriate oxidised and reduced form of a redox system. The metal merely acts as electrical contact. The potential arises due to the tendency of one form to change into the other form, e.g. $\mathrm{Pt/Fe^{2+}, Fe^{3+}}$, $\mathrm{Pt/Ce^{3+}, Ce^{4+}}$, $\mathrm{Pt/Sn^{2+}, Sn^{4+}}$. S003·7 · T003 00:11:52

Quinhydrone electrode. An inert metal such as platinum immersed in a solution containing quinone and hydroquinone; the metal merely acts as electrical contact and the potential arises due to the tendency of quinone to change to hydroquinone. Representation $\mathrm{Pt/Q, QH_2}$. S003·8 · T003 00:14:02

Quinhydrone. An equimolar (1 : 1) compound of quinone and hydroquinone; dissolving it gives $[Q] = [\mathrm{QH_2}]$, which is why the electrode's potential then depends only on pH. added for clarity S003·8

Statements in the class material

> *"The metal merely acts as electrical contact. The potential arises due to the tendency of one form to change in to other form."* S003·7

Key points

1. Both oxidised and reduced forms must be present in the same solution; Pt picks up the electrons exchanged between them. ESA ★★ T003 00:12:19 · T003 00:12:36
2. Unlike the gas electrode, no species is a gas and nothing is bubbled: both forms are ions in solution. ESA ★★ added for clarity S003·5 · S003·7
3. Quinhydrone: $\mathrm{Q + 2H^+ + 2e^- \rightleftharpoons QH_2}$, $E^\circ = +0.699$ V; with $[Q] = [QH_2]$, $E = 0.699 - 0.0591\,\mathrm{pH}$, so it measures pH. ESA ★★ S003·8 · T003 00:14:35
4. The quinhydrone electrode is not used above about pH 8, where hydroquinone (a weak acid) ionises and is oxidised by air. ESA ★ added for clarity S003·8

Why?

Why does the quinhydrone potential depend on pH, while $\mathrm{Pt|Sn^{4+}, Sn^{2+}}$ does not?
$\mathrm{H^+}$ takes part in the quinone reaction ($\mathrm{Q + 2H^+ + 2e^- \rightleftharpoons QH_2}$), so it appears in the Nernst equation; the tin reaction involves no $\mathrm{H^+}$.

S003·8 · T003 00:14:35

Exam traps

Exam trap. In the redox Nernst equation the reduced form goes on top: $\log \frac{[\mathrm{Sn^{2+}}]}{[\mathrm{Sn^{4+}}]}$. Inverting it flips the sign of the correction. S003·7 · T003 00:13:49

Exam trap. The gas and redox electrodes both use inert Pt; in a 2-mark note on either, say what Pt does (contact only) and what sets the potential (gas–ion equilibrium vs the ratio of the two ionic forms). ADDED FOR CLARITY ESA Dec-23 2d

7 Ion-selective (membrane) electrode

Definitions

Ion-selective electrode (membrane electrode). It consists of a membrane in contact with a solution, with which it can exchange ions; it develops a potential that depends on the concentration of one particular ion in the presence of other ions. E.g. the glass electrode, selective to $\mathbf{H^+}$, $\mathbf{Na^+}$, $\mathbf{K^+}$. S003·9 · T003 00:16:05 · PDF p.44 · ESA Jun-24a 2b · ESA Jan-25 s1 2b

Statements in the class material

> *"Equation for determining potential for pH sensitive Glass electrode:* $E_G = E_G^0 + 0.0591 \log_{10}[\mathbf{H^+}]$*."* S003·9

Key points

1. The membrane is the most important part: it exchanges ions with the solution and so develops a potential that depends on the concentration of one ion, even in the presence of other ions. ESA ★★ T003 00:16:43 · T003 00:18:14
2. Glass electrode: used in pH meters, selective to $\mathbf{H^+}$ (sometimes also $\mathbf{Na^+}$, $\mathbf{K^+}$). ESA ★★ S003·9 · T003 00:17:07
3. Other ISEs (older slides): crystalline $\mathbf{LaF_3}$ for $\mathbf{F^-}$, $\mathbf{Ag_2S}$ for $\mathbf{S^{2-}}$; liquid-membrane and PVC-immobilised ion exchangers for $\mathbf{Ca^{2+}}$, $\mathbf{K^+}$, $\mathbf{NO_3^-}$. ESA ★ (not on current slides) PDF p.45
4. A glass electrode cannot be used alone: it is coupled with an external reference such as the saturated calomel electrode, and the emf of that cell gives the pH (Unit 3). ESA ★★ (not on current slides) PDF p.57 · ESA Jun-24a 2b · ESA Jan-25 s1 2b

Why?

Why is it called ion-selective?
The membrane exchanges one particular ion, so the electrode responds to that ion's concentration and ignores (mostly) the others.
T003 00:16:05 · T003 00:16:25

Exam traps

Exam trap. Slide form $E_G = E_G^\circ + 0.0591 \log[\mathbf{H^+}]$ and pH form $E_G^\circ - 0.0591\,\mathbf{pH}$ are the same; mixing the sign of the pH term is the commonest error in the Unit 3 numericals. S003·9 · PDF p.56

Exam trap. In the glass + calomel cell, write E_{cell} to match your diagram: calomel on the left gives $E_{cell} = E_G - E_{cal}$ (PESU slides); glass on the left gives $E_{cell} = E_{cal} - E_G$. (added for clarity) PDF p.57

★ Past-paper questions

Ten deduplicated ESA papers (Jul 2023 a/b, Dec 2023, Jan 2024 set 2, May 2024, Jun 2024 a/b, Dec 2024 UE23 and UE22, Jan 2025 sets 1 and 2) were checked. Class 003 is asked three times: Dec 2023 Q2.d (4 marks: notes on the oxidation–reduction electrode and the gas electrode), and Jun 2024a Q2.b(i) and Jan 2025 set 1 Q2.b(i) (define an ion-selective electrode and write the cell when a glass electrode is coupled with a (saturated) calomel electrode). The glass + SCE pH numerical that follows those questions was asked in 9 of the 10 papers, but in the 2026 syllabus it belongs to Unit 3 (pH sensor, class 36). The metal–metal ion, metal–insoluble salt, amalgam and quinhydrone electrodes were never asked on their own.

6 · Oxidation–reduction electrodes and the quinhydrone electrode

ESA Dec 2023 (4 marks) UE22CY151B/A set 1, Q2.d · solved in W6.4

Write a note on the following types of electrodes: i. Oxidation-reduction electrodes (2M) ii. Gas electrode (2M).

7 · Ion-selective (membrane) electrode

ESA Jun 2024 (a) (2 marks) UE23CY151B/A set 1, Q2.b(i) · solved in W7.1

What is ion selective electrode? Write the cell representation when a glass electrode is coupled with a calomel electrode. (Part (ii), the pH numerical, is Unit 3.)

ESA Jan 2025 (2 marks) UE24CY151A set 1, Q2.b(i) · solved in W7.1

What is ion selective electrode? Write the cell representation when a glass electrode is coupled with a saturated calomel electrode. (Part (ii), the pH numerical, is Unit 3.)